

Where Should Machine Learning Go in Hybrid Trading Systems?

Matched-Budget Component Attribution Across Five Strategies and 153 Instruments

Version 5 — pre-registered, fully pipeline-generated results; code, data, and tests:
https://github.com/Cannonbold2412/ML_Component_Attribution

Abstract

Hybrid trading systems combine rule-based signals with machine-learned components, but rarely ask at which decision point ML adds value. We introduce Matched-Budget Component Attribution (MBCA): hold data, features, model class, hyperparameters, freeze discipline, and evaluation fixed, and replace one insertion point at a time — regime filter, meta-labeling, position sizing, entry, or exit — with a matched-budget learned counterpart. Study 1 applies MBCA to one strategy in three markets (0 significant improvements, 20 significant losses in 60 configurations). Study 2, pre-registered in version control before any data existed, applies it to five historically documented strategies (JMA and dual moving-average crossovers, Donchian breakouts, time-series momentum, RSI(2) mean reversion) on 153 daily series across 8 asset-class markets from 2000-01-03 onward, with two independent freeze dates, close-price stop fills, per-market costs, mark-to-market portfolios, a paired block bootstrap, and false-discovery-rate control. Of 528 market-level tests, 99 have raw $p < 0.05$ but only 8 survive as significant gains and 10 as losses. Narrowing components (meta-labeling, sizing) improve Sharpe in 67.6% and 70.7% of cells, are never significantly harmful, and transfer across markets, but their effect is small and replicates only partially across freezes. Replacing components are strategy-dependent: ML entry is the most harmful component, while ML exit is the only one that significantly improves a strategy in several markets (RSI(2), 6 significant gains). Execution realism dominates: stop-level fills inflate the rule-based controls' Sharpe by +0.34–+0.77. Every table, figure, and number is generated from the released pipeline.

Keywords: matched-budget component attribution, hybrid trading systems, meta-labeling, pre-registration, reproducibility, walk-forward validation, block bootstrap, false discovery rate, technical trading rules

Introduction

Much “ML for trading” work reports one hybrid configuration — a rule-based signal gated, filtered, or replaced by a learned model — against a baseline, and treats the Sharpe difference as evidence about machine learning in trading generally. This conflates *does ML help* with *does ML help at this specific decision point, with this feature set and model, holding everything else fixed*. A hybrid pipeline has several natural insertion points — which bars are admissible (regime filter), whether a signal-fired trade should be taken (meta-labeling), how much to risk (position sizing), where to enter, and when to exit — and a single aggregate comparison cannot say which of them is responsible for an improvement or a loss.

We introduce **Matched-Budget Component Attribution (MBCA)**: hold the data, features, model class and hyperparameters, freeze discipline, and evaluation protocol fixed, and vary only which insertion point a learned model replaces. We apply it twice. **Study 1** applies MBCA to one validated strategy (a JMA crossover) in three markets and finds 0 significant improvements in 60 configurations; an independent re-implementation reproduces that headline but not its market-level claims. **Study 2** is a pre-registered test of whether Study 1 generalizes: five historically documented strategies (JMA crossover, dual moving-average crossover, Donchian breakout, time-series momentum, RSI(2) mean reversion) on 153 instruments across 8 markets, two independent freeze dates, conservative execution, mark-to-market accounting, a paired block-

bootstrap test with false-discovery-rate control, and a battery of generalization, robustness, and bias tests.

Findings. (i) ML rarely changes a historically successful rule significantly once multiplicity is controlled: 8 significant gains and 10 significant losses in 528 market-level tests. (ii) Narrowing components (meta-labeling, position sizing) are the only ones that are directionally positive in most cells and never significantly harmful, and their effect transfers across markets — but it is small and replicates only partially across freezes. (iii) Replacing components are strategy-dependent: ML entry is the most harmful component, while ML exit, negative on average for trend-followers, is the only component that significantly improves a strategy in several markets (RSI(2) mean reversion). (iv) Execution assumptions dominate: filling stops at the stop level rather than at the breaching close inflates the rule-based control's Sharpe by +0.34–+0.77 and made ML exit look worse in Study 1 than it is. (v) The controls themselves lose most of their in-sample edge out of sample.

Contributions. MBCA and its taxonomy of insertion points; a pre-registered, auditable Study 2 in which data provenance, code, experiments, and every table, figure, and in-text number are generated by one pipeline from one configuration file; the component-level evidence above; and an open-source implementation in which a practitioner can plug in their own strategy and data.

Research Gap and Related Work

Ablation studies of ML-augmented trading pipelines exist (e.g. a 2025 cross-sectional A-share turnover pipeline ablates its component modules and attributes annualized-return contribution per module; discussed below), but they typically ablate *modules of one specific proprietary pipeline architecture*, not *insertion points of a single control mechanism into an already-fixed rule-based strategy* with identical features/model/hyperparameters held across every component. No work we found formalizes this constraint as a named, reusable methodology, runs the same feature set and model class through five distinct insertion points of one rule-based strategy with a uniform freeze/holdout/forward-tail protocol and bootstrap-CI significance testing at every insertion point, and reports the full grid of results — including the (numerous) negative ones — as a decision-support artifact rather than the single configuration that happened to look best. MBCA fills that gap: it is the methodology, not merely the experiment, that is missing from the literature.

Meta-labeling. López de Prado’s meta-labeling [1] separates the “which direction” decision (the primary model) from “should I take this specific signal” (a secondary classifier trained on realized outcomes) — exactly the mechanism this paper’s ML Meta-Labeling component implements, though the primary model here is a fixed rule (JMA crossover), not a learned one. In MBCA’s taxonomy (Section 3.2), meta-labeling is a canonical *narrowing* component: it can only reject decisions the primary system already made, never invent new ones.

Regime-conditioned and hybrid rule+ML systems. A 2026 hybrid system integrating technical rules, ML, and financial sentiment for regime-adaptive equity strategies [2] reports aggregate improvements over a rule-only baseline but does not decompose which sub-component is responsible — precisely the conflation MBCA is designed to resolve. A 2025 deep-learning-enhanced, multi-day-turnover A-share pipeline [3] is the closest methodological precedent we found: it ablates its five pipeline modules one at a time and reports each module’s marginal contribution to annualized return — the same spirit as MBCA’s attribution table, but over a different, proprietary module set, without a uniform ML feature/model budget held across modules, without a named, reusable methodology, and without holdout-to-forward-tail generalization testing per module.

Position sizing and volatility-managed portfolios. Moreira and Muir’s volatility-managed portfolios [4] scale exposure by a *realized-volatility* signal at the portfolio level; this paper’s ML Position Sizing component instead scales exposure by a *predicted-win-probability* signal at the individual-trade level — another *narrowing* component in MBCA’s taxonomy, since it re-weights rather than rejects.

Time-series momentum. Moskowitz, Ooi and Pedersen [5] is the standard reference for the rule-based trend-

following family this paper’s baseline belongs to; our contribution is orthogonal to theirs.

Technical trading rules and data snooping. Simple moving-average and breakout rules have a long empirical record [6], but their apparent profitability is highly sensitive to how many rules were searched: [7] show that a bootstrap accounting for the full rule universe removes much of it, and [8] argue for multiple-testing thresholds far above conventional ones. Study 2 therefore evaluates canonical, pre-registered rules rather than searched ones, compares Sharpe ratios with a dependence-robust paired bootstrap [9], and controls the false-discovery rate across every test it runs.

Gap. No work we found holds a rule-based strategy, its data, its features, its model class, and its evaluation protocol fixed while sweeping which single component ML replaces, names and formalizes that constraint as a reusable methodology, and reports every resulting configuration’s forward-tested significance rather than the best one. MBCA is that methodology; this paper is its first full application.

Methodology: Matched-Budget Component Attribution

Definition

MBCA is a controlled-variable methodology for decomposing a hybrid decision system’s performance by insertion point. It applies whenever a system combines a validated, rule-based control with one or more points where a learned model could instead make the decision. MBCA holds five things fixed across every tested insertion point:

- the **data** each component’s model is trained and evaluated on;
- the **feature set** available to every learned component;
- the **model class and hyperparameters** (no per-component tuning);
- the **training/freeze discipline** (fit once, never refit on evaluation data); and
- the **evaluation protocol and statistical test** used to call a result significant.

Only the *insertion point* — which single decision in the pipeline the learned model replaces or modifies — varies between configurations, so a performance difference can be attributed to the insertion point rather than to a confound of feature engineering, tuning budget, or evaluation leniency. A negative result under MBCA is a statement about *this stated budget* at that insertion point, not a claim that no learned model could ever help there.

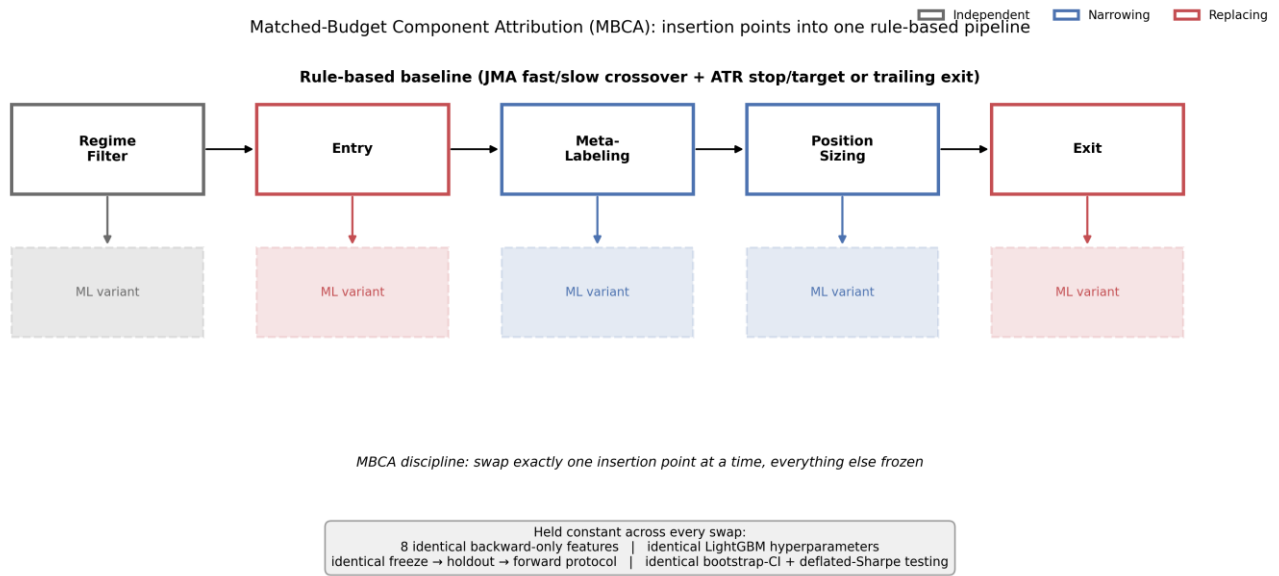
A taxonomy of insertion points

We classify each component by how much it can alter the trade set the rule-based system would otherwise produce:

- **Independent** — gates admissibility orthogonally to the entry signal; can only reject bars. *Tested*: ML regime filter (and, as a rule-based comparator, an $ADX \geq 25$ gate).
- **Narrowing** — filters or re-weights decisions the rule-based system already made; cannot invent trades. *Tested*: meta-labeling, position sizing.

- **Replacing** — substitutes the rule-based mechanism outright; can generate a materially different trade set. *Tested*: entry generation, exit policy.

The ex-ante hypothesis is that narrowing components bound their own worst case and should show gentler, lower-variance effects than replacing components.



The MBCA experiment: insertion points into one rule-based pipeline, each with a matched-budget ML counterpart, and the invariants held constant across every swap.

Shared feature set, model, and freeze

Every learned component uses the same 8 backward-only features — ADX, ATR-normalized momentum, historical volatility, return z-score, backward-looking R^2 , RSI, MACD histogram, and KAMA slope — each computable from data up to and including bar t . Every classifier is a `lightgbm.LGBMClassifier` with identical hyperparameters (`n_estimators=200`, `num_leaves=31`, `max_depth=6`, `learning_rate=0.05`, `min_child_samples=30`, `class_weight="balanced"`, `random_state=42`) on standardized features. Each classifier is fit exactly once on data before a freeze date; training rows whose label window crosses the freeze are excluded; after freezing, models only predict. Parameters of the rule-based part of every pipeline (signal lengths, stop multiples) are re-selected by the same anchored walk-forward harness for the control and for every variant: each instrument's history is cut into five equal chronological folds, the parameter set with the best mean in-sample trade return (subject to a minimum trade count) is chosen on fold k and traded on fold $k + 1$.

The components

For bar t of instrument i let $\mathbf{x}_{i,t} \in \mathbb{R}^8$ be the shared feature vector.

- **ML regime filter**: entries are allowed iff $P(\text{trend}_{i,t} = 1 \mid \mathbf{x}_{i,t}) \geq 0.5$, where the label is 1 when the next 15 closes fit a straight line with $R^2 \geq 0.5$ (“linearity”; Study 1 also tests two other label definitions).
- **ML meta-labeling**: at bars where the rule-based entry fires, the label is whether the trade the control would open is profitable; trades are taken iff $\hat{p}_{i,t}^{\text{meta}} \geq 0.5$ [1].
- **ML position sizing**: the same frozen meta-label model, consumed continuously: every trade is sized $m_{i,t} = 2\hat{p}_{i,t}^{\text{meta}} \in [0, 2]$ ($p = 0.5 \mapsto 1 \times$); no trade is ever added or removed.
- **ML entry**: two classifiers estimate whether a long (short) bracket entered at t with stop 1.5 ATR and target 3 ATR hits the target first within 40 bars; the signal is the direction whose probability is larger and ≥ 0.5 . This replaces the control's entry rule.
- **ML exit**: two classifiers estimate whether an open long (short) position is still favourable 10 bars ahead; the position exits at the first bar where that probability falls below 0.5, or at a 3 ATR safety stop. This replaces the control's exit rule.
- **Meta-label + sizing (“Best Combined”)**: meta-labeling filters, sizing re-weights the survivors.

In Study 1 this pairing was chosen after the fact; in Study 2 it was *pre-registered*, so its Study 2 result is an out-of-sample test of that choice.

Study 1: One Strategy, Three Markets

Design

The control is a JMA(7)/JMA(21) crossover whose parameters are walk-forward selected from a small grid, with a fixed $1.5 \times \text{ATR}$ stop / $3 \times \text{ATR}$ target (forex) or a $1.5 \times \text{ATR}$ stop with a ratcheting trailing stop (commodities, Indian equities). Markets: a 4-pair forex sleeve (AUDNZD, CADJPY, AUDUSD, NZDUSD), five commodities (WTI, Brent, natural gas, gold, silver), and 48 NIFTY-50 stocks with an annual sector-curation overlay. Each market's models were frozen at 70% of its calendar span; results are reported on a *holdout* window after the freeze and, for commodities and Indian equities, on a *forward* window of data fetched after the study was designed. Study 1 reports a trade stream as a daily series by averaging the returns of trades exiting each day, Sharpe ratios annualized with a 6%/yr risk-free rate, and a

95% CI from 2,000 *unpaired* i.i.d. bootstrap resamples of each variant's daily series, computed on the *per-day* Sharpe difference (so CI bounds are $\sqrt{252} \approx 15.9$ times smaller than the annualized ΔSharpe beside them; Appendix 11). A configuration is significant iff this CI excludes zero.

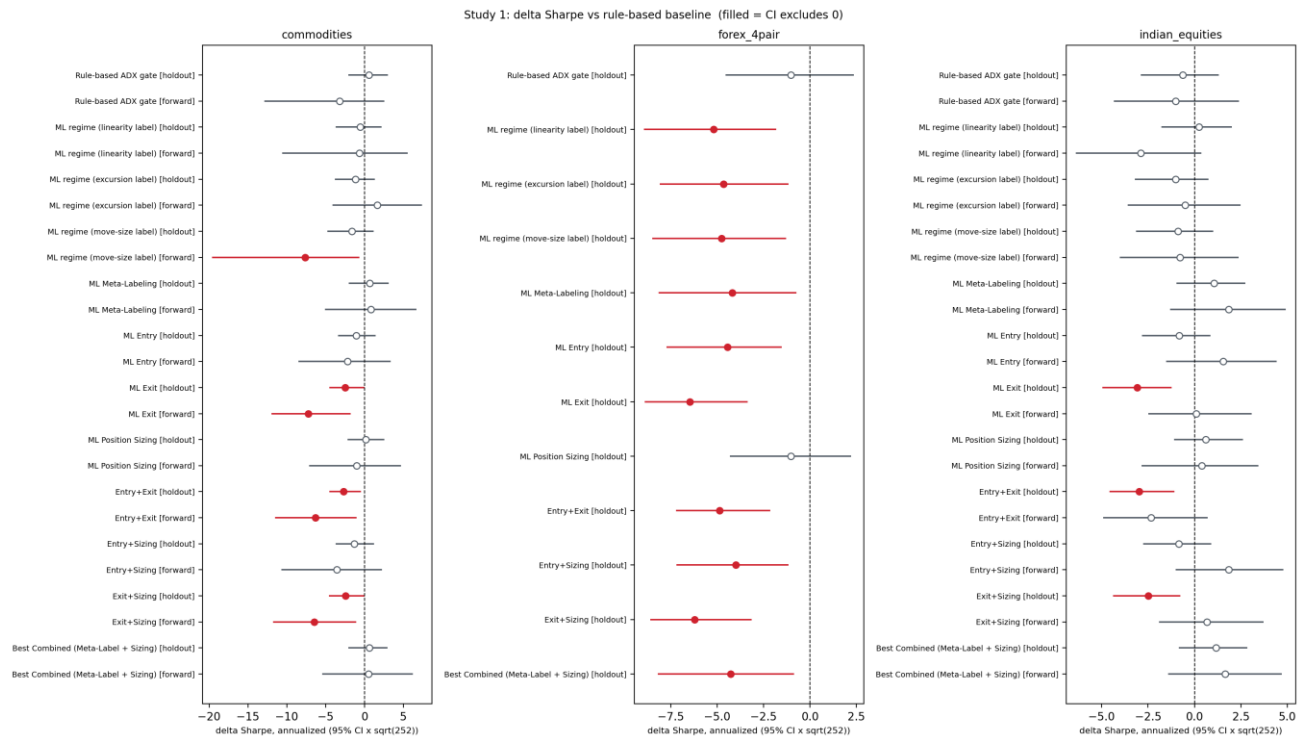
Results

Across 60 configurations (5 components, 4 combinations, 3 markets, up to 2 windows), **0 show a significant improvement and 20 a significant loss** (Table 1, Figures 2–3). ML exit is the clearest failure: it loses significantly in 4 of its 5 configurations and inflates trade count 2.5–4.2×, because a per-bar continuation classifier exits early and un-blocks crossovers that a longer-held rule-based trade would have absorbed. On the original forex data, where the control's holdout Sharpe is 5.53, 10 of 12 configurations lose significantly. The closest call is meta-labeling followed by position sizing in Indian equities: holdout Sharpe 2.84→4.01 (CI [−0.050,0.176] per day, roughly [−0.8,2.8] annualized) and forward 1.05→2.70 (CI [−0.087,0.293]) — not significant.

Study 1: ML Opportunity Matrix (cell = mean delta Sharpe over windows; * = CI excludes 0; label colour = independent / narrowing / replacing / combination)

	commodities	forex_4pair	indian_equities	
Rule-based ADX gate	-1.34	-1.02	-0.82	Neutral (no consistent effect)
ML regime (linearity label)	-0.61	-5.20 *	-1.33	Avoid (significant loss in forex_4pair)
ML regime (excursion label)	+0.23	-4.66 *	-0.76	Mixed (avoid in forex_4pair; promising, unreplicated in commodities)
ML regime (move-size label)	-4.65 *	-4.78 *	-0.84	Avoid (significant loss in commodities, forex_4pair)
ML Meta-Labeling	+0.75	-4.19 *	+1.44	Mixed (avoid in forex_4pair; promising, unreplicated in commodities, indian_equities)
ML Entry	-1.61	-4.46 *	+0.36	Mixed (avoid in forex_4pair; promising, unreplicated in indian_equities)
ML Exit	-4.84 *	-6.48 *	-1.49 *	Avoid (significant loss in commodities, forex_4pair, indian_equities)
ML Position Sizing	-0.44	-1.03	+0.49	Worth exploring in indian_equities (unreplicated)
Entry+Exit	-4.49 *	-4.88 *	-2.65 *	Avoid (significant loss in commodities, forex_4pair, indian_equities)
Entry+Sizing	-2.44	-3.99 *	+0.50	Mixed (avoid in forex_4pair; promising, unreplicated in indian_equities)
Exit+Sizing	-4.43 *	-6.21 *	-0.90 *	Avoid (significant loss in commodities, forex_4pair, indian_equities)
Best Combined (Meta-Label + Sizing)	+0.55	-4.27 *	+1.40	Mixed (avoid in forex_4pair; promising, unreplicated in commodities, indian_equities)

Study 1 ML Opportunity Matrix: mean ΔSharpe per component and market (* = CI excludes zero) with the mechanically derived recommendation.



Study 1, all configurations. Dots: annualized Δ Sharpe; intervals: per-day bootstrap CIs rescaled by $\sqrt{252}$ so that both share one axis. Filled = CI excludes zero.

Independent re-implementation

The released package re-implements Study 1 from scratch (Appendix 10); running it on the bundled data is an independent replication with three known differences (a different forex data vendor with a flat 7 bps cost, no sector-curation overlay, one walk-forward over the full history). It reproduces the headline — 0 significant gains, ML exit significantly harmful in every market with trade count inflated 2.4–3.0 $\times$, 49 of 60 configurations agreeing in sign — but not the market-level claims: with different forex data (control Sharpe 1.39) only 2 forex configurations lose significantly and meta-labeling flips

Study 1: all 60 configurations (original study results). CI = unpaired i.i.d. bootstrap of the per-day Sharpe difference.

from -4.19 to +0.94; and the Best Combined near-miss replicates on the forward window (+1.65 vs. +1.53) but not on the holdout (+1.16 vs. +0.04), where the control itself rises to 4.31 without curation (Table 2).

Study 1 therefore leaves three questions open, which Study 2 was pre-registered to answer: does the pattern hold (i) for other historically successful strategies, not only a JMA crossover, (ii) across a much broader and more diverse universe, and (iii) at a second, independent freeze date, under stricter execution assumptions and a test that respects the pairing and serial dependence of returns?

Component	Variant	Market	Window	Base	Variant	Δ SR	95% CI (per day)	Sig.
ML Regime Filter	Rule-based ADX gate	commodities	forward	4.17	0.94	-3.23	[-0.807, 0.153]	no
ML Regime Filter	ML regime (linearity label)	commodities	forward	4.17	3.50	-0.67	[-0.662, 0.345]	no
ML Regime Filter	ML regime (excursion label)	commodities	forward	4.17	5.80	+1.63	[-0.254, 0.461]	no
ML Regime Filter	ML regime (move-size label)	commodities	forward	4.17	-3.48	-7.65	[-1.229, -0.048]	yes
ML Regime Filter	Rule-based ADX gate	commodities	holdout	2.72	3.27	+0.54	[-0.126, 0.184]	no
ML Regime Filter	ML regime (linearity label)	commodities	holdout	2.72	2.18	-0.54	[-0.229, 0.131]	no
ML Regime Filter	ML regime (excursion label)	commodities	holdout	2.72	1.57	-1.16	[-0.233, 0.076]	no
ML Regime Filter	ML regime (move-size label)	commodities	holdout	2.72	1.08	-1.65	[-0.296, 0.069]	no

Component	Variant	Market	Window	Base	Variant	Δ SR	95% CI (per day)	Sig.
ML Regime Filter	Rule-based ADX gate	forex_4pair	holdout	5.53	4.51	-1.02	[-0.284, 0.147]	no
ML Regime Filter	ML regime (linearity label)	forex_4pair	holdout	5.53	0.33	-5.20	[-0.561, -0.117]	yes
ML Regime Filter	ML regime (excursion label)	forex_4pair	holdout	5.53	0.87	-4.66	[-0.507, -0.076]	yes
ML Regime Filter	ML regime (move-size label)	forex_4pair	holdout	5.53	0.75	-4.78	[-0.533, -0.084]	yes
ML Regime Filter	Rule-based ADX gate	indian_equities	forward	2.50	1.48	-1.01	[-0.270, 0.148]	no
ML Regime Filter	ML regime (linearity label)	indian_equities	forward	2.50	-0.40	-2.89	[-0.399, 0.021]	no
ML Regime Filter	ML regime (excursion label)	indian_equities	forward	2.50	1.98	-0.51	[-0.224, 0.154]	no
ML Regime Filter	ML regime (move-size label)	indian_equities	forward	2.50	1.71	-0.78	[-0.250, 0.147]	no
ML Regime Filter	Rule-based ADX gate	indian_equities	holdout	2.84	2.21	-0.63	[-0.179, 0.080]	no
ML Regime Filter	ML regime (linearity label)	indian_equities	holdout	2.84	3.08	+0.23	[-0.109, 0.124]	no
ML Regime Filter	ML regime (excursion label)	indian_equities	holdout	2.84	1.83	-1.01	[-0.200, 0.045]	no
ML Regime Filter	ML regime (move-size label)	indian_equities	holdout	2.84	1.95	-0.89	[-0.195, 0.061]	no
ML Meta-Labeling	ML meta-label filter	commodities	forward	4.17	5.01	+0.83	[-0.313, 0.414]	no
ML Meta-Labeling	ML meta-label filter	commodities	holdout	2.72	3.39	+0.67	[-0.123, 0.190]	no
ML Meta-Labeling	ML meta-label filter	forex_4pair	holdout	5.53	1.34	-4.19	[-0.512, -0.049]	yes
ML Meta-Labeling	ML meta-label filter	indian_equities	forward	1.05	2.90	+1.84	[-0.080, 0.307]	no
ML Meta-Labeling	ML meta-label filter	indian_equities	holdout	2.84	3.89	+1.04	[-0.059, 0.169]	no
ML Entry	ML entry (replaces jma_signals)	commodities	forward	4.17	2.00	-2.17	[-0.530, 0.206]	no
ML Entry	ML entry (replaces jma_signals)	commodities	holdout	2.72	1.66	-1.06	[-0.210, 0.083]	no
ML Entry	ML entry (replaces jma_signals)	forex_4pair	holdout	5.53	1.07	-4.46	[-0.484, -0.099]	yes
ML Entry	ML entry (replaces jma_signals)	indian_equities	forward	1.05	2.59	+1.53	[-0.093, 0.275]	no
ML Entry	ML entry (replaces jma_signals)	indian_equities	holdout	2.84	2.02	-0.82	[-0.175, 0.051]	no
ML Exit	ML exit (replaces ATR stop/target)	commodities	forward	4.17	-3.04	-7.22	[-0.749, -0.119]	yes
ML Exit	ML exit (replaces ATR stop/target)	commodities	holdout	2.72	0.25	-2.47	[-0.278, -0.005]	yes
ML Exit	ML exit (replaces ATR stop/target)	forex_4pair	holdout	5.53	-0.95	-6.48	[-0.558, -0.214]	yes
ML Exit	ML exit (replaces ATR stop/target)	indian_equities	forward	1.05	1.14	+0.09	[-0.154, 0.191]	no
ML Exit	ML exit (replaces ATR	indian_equities	holdout	2.84	-0.24	-3.08	[-0.310, -	yes

Component	Variant	Market	Window	Base	Variant	Δ SR	95% CI (per day)	Sig.
	stop/target)						0.080]	
ML Position Sizing	ML position sizing (re-weighted trades)	commodities	forward	4.17	3.14	-1.04	[-0.444, 0.289]	no
ML Position Sizing	ML position sizing (re-weighted trades)	commodities	holdout	2.72	2.89	+0.16	[-0.132, 0.153]	no
ML Position Sizing	ML position sizing (re-weighted trades)	forex_4pair	holdout	5.53	4.51	-1.03	[-0.269, 0.137]	no
ML Position Sizing	ML position sizing (re-weighted trades)	indian_equities	forward	1.05	1.43	+0.38	[-0.177, 0.214]	no
ML Position Sizing	ML position sizing (re-weighted trades)	indian_equities	holdout	2.84	3.44	+0.60	[-0.067, 0.162]	no
Combinations	Entry+Exit	commodities	forward	4.17	-2.14	-6.31	[-0.718, -0.070]	yes
Combinations	Entry+Sizing	commodities	forward	4.17	0.62	-3.55	[-0.667, 0.136]	no
Combinations	Exit+Sizing	commodities	forward	4.17	-2.28	-6.45	[-0.736, -0.072]	yes
Combinations	Best Combined (Meta-Label + Sizing)	commodities	forward	4.17	4.67	+0.49	[-0.337, 0.385]	no
Combinations	Entry+Exit	commodities	holdout	2.72	0.05	-2.68	[-0.278, -0.036]	yes
Combinations	Entry+Sizing	commodities	holdout	2.72	1.40	-1.33	[-0.227, 0.071]	no
Combinations	Exit+Sizing	commodities	holdout	2.72	0.30	-2.42	[-0.281, -0.006]	yes
Combinations	Best Combined (Meta-Label + Sizing)	commodities	holdout	2.72	3.34	+0.61	[-0.125, 0.181]	no
Combinations	Entry+Exit	forex_4pair	holdout	5.53	0.66	-4.88	[-0.453, -0.137]	yes
Combinations	Entry+Sizing	forex_4pair	holdout	5.53	1.54	-3.99	[-0.451, -0.076]	yes
Combinations	Exit+Sizing	forex_4pair	holdout	5.53	-0.68	-6.21	[-0.540, -0.201]	yes
Combinations	Best Combined (Meta-Label + Sizing)	forex_4pair	holdout	5.53	1.26	-4.27	[-0.514, -0.057]	yes
Combinations	Entry+Exit	indian_equities	forward	1.05	-1.29	-2.34	[-0.307, 0.042]	no
Combinations	Entry+Sizing	indian_equities	forward	1.05	2.89	+1.83	[-0.061, 0.299]	no
Combinations	Exit+Sizing	indian_equities	forward	1.05	1.73	+0.68	[-0.117, 0.231]	no
Combinations	Best Combined (Meta-Label + Sizing)	indian_equities	forward	1.05	2.70	+1.65	[-0.087, 0.293]	no
Combinations	Entry+Exit	indian_equities	holdout	2.84	-0.13	-2.97	[-0.285, -0.071]	yes
Combinations	Entry+Sizing	indian_equities	holdout	2.84	2.00	-0.84	[-0.171, 0.054]	no
Combinations	Exit+Sizing	indian_equities	holdout	2.84	0.36	-2.48	[-0.273, -0.051]	yes
Combinations	Best Combined (Meta-Label + Sizing)	indian_equities	holdout	2.84	4.01	+1.16	[-0.050, 0.176]	no

Study 1: original results vs. independent re-implementation (annualized Δ Sharpe).

Market	Window	Variant	Original	Replication	Orig.	Repl.
commodities	forward	Rule-based ADX gate	-3.23	-5.03	n.s.	n.s.
commodities	forward	ML Regime Filter (linearity)	-0.67	-3.51	n.s.	n.s.
commodities	forward	ML Regime Filter (excursion)	+1.63	+2.13	n.s.	n.s.
commodities	forward	ML Regime Filter (move_size)	-7.65	+1.63	sig.	n.s.
commodities	holdout	Rule-based ADX gate	+0.54	+0.51	n.s.	n.s.
commodities	holdout	ML Regime Filter (linearity)	-0.54	+2.19	n.s.	n.s.
commodities	holdout	ML Regime Filter (excursion)	-1.16	+0.20	n.s.	n.s.
commodities	holdout	ML Regime Filter (move_size)	-1.65	-0.81	n.s.	n.s.
forex	holdout	Rule-based ADX gate	-1.02	-1.05	n.s.	n.s.
forex	holdout	ML Regime Filter (linearity)	-5.20	-0.02	sig.	n.s.
forex	holdout	ML Regime Filter (excursion)	-4.66	-3.55	sig.	n.s.
forex	holdout	ML Regime Filter (move_size)	-4.78	-2.16	sig.	n.s.
indian_equities	forward	Rule-based ADX gate	-1.01	-0.43	n.s.	n.s.
indian_equities	forward	ML Regime Filter (linearity)	-2.89	-0.21	n.s.	n.s.
indian_equities	forward	ML Regime Filter (excursion)	-0.51	-0.43	n.s.	n.s.
indian_equities	forward	ML Regime Filter (move_size)	-0.78	+0.29	n.s.	n.s.
indian_equities	holdout	Rule-based ADX gate	-0.63	-1.39	n.s.	n.s.
indian_equities	holdout	ML Regime Filter (linearity)	+0.23	-1.35	n.s.	n.s.
indian_equities	holdout	ML Regime Filter (excursion)	-1.01	-1.27	n.s.	n.s.
indian_equities	holdout	ML Regime Filter (move_size)	-0.89	-1.69	n.s.	sig.
commodities	forward	ML Meta-Labeling	+0.83	-0.47	n.s.	n.s.
commodities	holdout	ML Meta-Labeling	+0.67	+1.27	n.s.	n.s.
forex	holdout	ML Meta-Labeling	-4.19	+0.94	sig.	n.s.
indian_equities	forward	ML Meta-Labeling	+1.84	+1.40	n.s.	n.s.
indian_equities	holdout	ML Meta-Labeling	+1.04	-0.02	n.s.	n.s.
commodities	forward	ML Entry	-2.17	-3.50	n.s.	n.s.
commodities	holdout	ML Entry	-1.06	-0.12	n.s.	n.s.
forex	holdout	ML Entry	-4.46	-1.06	sig.	n.s.
indian_equities	forward	ML Entry	+1.53	+0.37	n.s.	n.s.
indian_equities	holdout	ML Entry	-0.82	-1.86	n.s.	sig.
commodities	forward	ML Exit	-7.22	-6.16	sig.	sig.
commodities	holdout	ML Exit	-2.47	-1.61	sig.	n.s.
forex	holdout	ML Exit	-6.48	-3.87	sig.	sig.
indian_equities	forward	ML Exit	+0.09	-0.12	n.s.	n.s.
indian_equities	holdout	ML Exit	-3.08	-5.07	sig.	sig.
commodities	forward	ML Position Sizing	-1.04	-1.13	n.s.	n.s.
commodities	holdout	ML Position Sizing	+0.16	+0.18	n.s.	n.s.
forex	holdout	ML Position Sizing	-1.03	-0.79	n.s.	n.s.
indian_equities	forward	ML Position Sizing	+0.38	+0.18	n.s.	n.s.
indian_equities	holdout	ML Position Sizing	+0.60	+0.16	n.s.	n.s.
commodities	forward	Entry+Exit	-6.31	-7.19	sig.	sig.
commodities	forward	Entry+Sizing	-3.55	-5.86	n.s.	sig.
commodities	forward	Exit+Sizing	-6.45	-5.45	sig.	sig.
commodities	forward	Best Combined (Meta-Label + Sizing)	+0.49	-0.93	n.s.	n.s.
commodities	holdout	Entry+Exit	-2.68	-1.66	sig.	n.s.
commodities	holdout	Entry+Sizing	-1.33	-0.34	n.s.	n.s.

Market	Window	Variant	Original	Replication	Orig.	Repl.
commodities	holdout	Exit+Sizing	-2.42	-1.31	sig.	n.s.
commodities	holdout	Best Combined (Meta-Label + Sizing)	+0.61	+1.19	n.s.	n.s.
forex	holdout	Entry+Exit	-4.88	-3.34	sig.	n.s.
forex	holdout	Entry+Sizing	-3.99	-0.93	sig.	n.s.
forex	holdout	Exit+Sizing	-6.21	-3.91	sig.	sig.
forex	holdout	Best Combined (Meta-Label + Sizing)	-4.27	+1.11	sig.	n.s.
indian_equities	forward	Entry+Exit	-2.34	-2.44	n.s.	n.s.
indian_equities	forward	Entry+Sizing	+1.83	+0.36	n.s.	n.s.
indian_equities	forward	Exit+Sizing	+0.68	+0.40	n.s.	n.s.
indian_equities	forward	Best Combined (Meta-Label + Sizing)	+1.65	+1.53	n.s.	n.s.
indian_equities	holdout	Entry+Exit	-2.97	-5.02	sig.	sig.
indian_equities	holdout	Entry+Sizing	-0.84	-2.13	n.s.	sig.
indian_equities	holdout	Exit+Sizing	-2.48	-4.39	sig.	sig.
indian_equities	holdout	Best Combined (Meta-Label + Sizing)	+1.16	+0.04	n.s.	n.s.

Data and provenance

Study 2: Five Historical Strategies Across a Broad Universe

Pre-registration and single source of truth

Every Study 2 design choice — universe, cleaning rules, strategies and parameter grids, components, freeze dates, execution model, statistics, regime definitions, and crisis windows — is declared in one configuration file (`study.toml`) that was committed to version control *before* any Study 2 data was downloaded or any result existed. The implementation reads nothing else: the same file drives data collection, every experiment, and every table, figure, and in-text number of this section and the next, which are generated by the pipeline rather than typed (Appendix 10). The only deviation from the pre-registration was forced by data availability and is logged: one planned symbol (Roche, `ROG.SW`) was not served by the data source, leaving 153 of 154 planned series. The regime-filter component was restricted in advance to a single label definition so that it could not be chosen after the fact.

Daily OHLC data for 153 instruments in 8 markets (Table 3) — 28 equity indices from the Americas, Europe, and Asia-Pacific, 19 US sector and country ETFs, 11 bond ETFs and Treasury futures, 19 commodity futures, 18 currency pairs including emerging-market currencies, 4 cryptocurrencies, and 20 US plus 34 non-US large-cap stocks on 14 exchanges — were downloaded from the Yahoo Finance chart API (fetched 2026-09-21, 2000-01-03 to 2026-09-21, 978,717 daily bars after cleaning). The raw API responses are stored byte-for-byte with their URL, UTC fetch time, and SHA-256 hash; cleaning is a deterministic function of the raw bytes and the declared rules (split/dividend adjustment via the adjusted close; removal of 7,890 rows with missing or non-positive prices; 10,868 OHLC-consistency repairs; 2 reversing bad ticks). Two independent checks validate the data: against 17 FRED series (5-day return correlation 0.82–1.00, median absolute price difference at most 1.2%; daily FX correlations are lower only because Yahoo and FRED fix FX at different times of day, and natural-gas front-month futures vs. Henry Hub spot, a different instrument, correlate at 0.64), and a browser spot-check of 40 closing prices for 2 symbols against MarketWatch (maximum difference 0.00).

Study 2 universe after cleaning (Yahoo Finance daily data, cross-checked against FRED and MarketWatch). Cost = round-trip transaction cost + slippage in bps.

Market	N	First	Last	Median rows	Cost (bps)	Survivorship-free
Equity indices	28	2000-01-03	2026-09-21	6,654	5	yes
Equity ETFs	19	2000-01-03	2026-09-21	6,719	5	yes
Fixed income	11	2000-09-21	2026-09-21	6,075	5	yes
Commodities	19	2000-01-03	2026-09-21	6,542	5	yes
FX	18	2000-01-03	2026-09-21	5,929	3	yes
Crypto	4	2014-09-17	2026-09-21	3,813	20	no
US stocks	20	2000-01-03	2026-09-21	6,719	10	no
Intl stocks	34	2000-01-03	2026-09-21	6,711	10	no

The five strategies

Five historically documented rule families serve as controls (Table 4): the JMA crossover of Study 1; the dual moving-average crossover of [6]; the Donchian channel breakout of the Turtle rules [10]; time-series

The five pre-registered strategies (the rule-based controls). Each instrument's parameters are re-selected by walk-forward on every fold.

Strategy	Reference	Signal grid	Exit	Exit grid
JMA crossover (paper baseline)	Study 1 baseline; Jurik moving average crossover	fast=[5, 7, 9], slow=[21, 28]	trailing	sl_mult=[1.5, 2.0], trail_mult=[2.0, 3.0]
Dual moving-average crossover	Brock, Lakonishok & LeBaron (1992)	fast=[20, 50], slow=[100, 200]	trailing	sl_mult=[2.0, 3.0], trail_mult=[3.0, 4.0]
Donchian channel breakout	Turtle trading rules (Dennis/Eckhardt); Donchian (1960)	lookback=[20, 55]	trailing	sl_mult=[2.0], trail_mult=[2.0, 3.0]
Time-series momentum	Moskowitz, Ooi & Pedersen (2012)	lookback=[126, 252]	trailing	sl_mult=[2.0, 3.0], trail_mult=[3.0, 4.0]
RSI(2) short-term mean reversion	Connors & Alvarez (2008)	entry=[5, 10]	bracket	sl_mult=[2.0, 3.0], tp_mult=[1.0, 1.5]

Execution and portfolio accounting

Execution is deliberately conservative. Signals and exits are evaluated on daily closes; a stop breached by a close is filled *at that close*, not at the stop level (the optimistic stop-level fill of Study 1 is re-run as a sensitivity), and entries fill at the signal bar's close (a one-day entry lag is re-run as a sensitivity). Round-trip transaction cost plus slippage is set per market in advance (3 bps FX, 5 bps indices/ETFs/futures, 10 bps single stocks, 20 bps crypto) and varied *imes0*, *imes2*, *imes4*. Performance is measured on *mark-to-market* daily portfolios: capital is split equally across the instruments of a market (or of the whole universe) that trade on a given day, each position earns its instrument's close-to-close return from entry to exit, and half the round-trip cost is charged at entry and half at exit, scaled by position size. We report CAGR, volatility, Sharpe and Sortino ratios (no risk-free rate, since the universe spans currencies), maximum drawdown, daily 95% CVaR, skewness, one-way turnover, and trade statistics.

Windows, generalization, and robustness design

Two independent freezes give two non-overlapping out-of-sample windows, each with its own frozen models: **F2008** (models trained on 2000–2007, tested 2008–2017, including the global financial crisis) and **F2018** (trained on 2000–2017, tested 2018 onwards, including the COVID crash and the 2022 bear market). Crypto enters only F2018. Each (strategy, market, freeze) comparison is a market-level test; the equal-weight portfolio of all 153 instruments gives a global test per (strategy, freeze). Beyond the main grid, the design pre-registers: *cross-market transfer* (meta-labeling and meta-label + sizing models fit on every *other* market, then applied to the held-out market); *budget ablations* (meta-label thresholds 0.45 and 0.55; a small model with 50 trees of depth 3 for meta-labeling and exit); *execution* sensitivities; cost multipliers; *cross-regime* tests (bull/bear by the market index's position vs. its 200-day average, high/low

momentum [5], entered when the sign of the trailing 6- or 12-month return flips; and RSI(2) short-term mean reversion in the direction of the 200-day trend [11]. Four are trend-following, one is contrarian, and their holding periods range from days to months. Each control's parameters are walk-forward selected per instrument exactly as in Study 1.

volatility by 20-day realized volatility vs. its expanding median, both known at $t - 1$); five crisis windows; a survivorship split (survivorship-free indices/ETFs/futures/FX vs. current-constituent stocks and surviving coins); leave-one-instrument-out and leave-one-market-out jackknives; and an in-sample vs. out-of-sample decay check of the controls' parameter selection.

Inference

The difference in annualized Sharpe ratio between a variant and its control is tested with a paired circular-block bootstrap [9], [12]: both daily return series are aligned on the same dates and the *same* 20-day blocks are resampled for both (2,000 replicates), preserving their cross-correlation and short-range serial dependence — correcting the unpaired i.i.d. test of Study 1. The two-sided p -value is $2\min\{P(\Delta^* \leq 0), P(\Delta^* \geq 0)\}$. Multiplicity is controlled with the Benjamini–Hochberg false-discovery rate [13] within pre-defined families (the 528 market-level main tests; the 70 global main tests; transfer; ablation; execution), with Holm's method [14] reported as a family-wise check, in the spirit of [8] and [7]. A difference is called significant iff $q < 0.05$. Deflated Sharpe ratios [15] are reported for every portfolio.

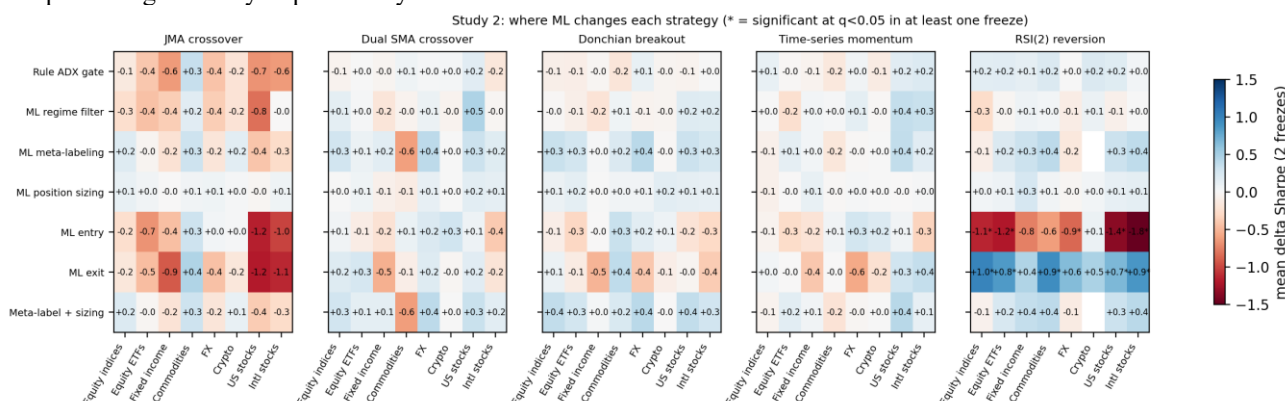
Study 2: Results

Headline: few significant effects, and they are strategy-specific

Of the 528 pre-registered market-level main tests (7 components $\times$ 5 strategies $\times$ 8 markets $\times$ 2 freezes, less cells without training data), 99 have a raw $p < 0.05$ — far more than the $\approx 5\%$ expected under a global null, so the ML components do change performance — but after Benjamini–Hochberg correction only 8 are significant gains and 10 significant losses (Holm: 16 of either sign). 54.9% of all market-level differences are positive. On the all-market portfolio, which diversifies across 153

instruments, 1 of the 70 global tests is a significant gain and 1 a significant loss (Table 7, Figure 5). No component significantly improves any of the four trend-

following strategies on the all-market portfolio in either freeze.



Mean Δ Sharpe over the two freezes for every strategy, component, and market (* = significant at $q < 0.05$ in at least one freeze).

Component attribution

Table 5 and Figure 4 give the per-component picture, which refines the taxonomy of Section 3.2:

- **Narrowing components are the most consistently positive and never significantly harmful.** Meta-labeling improves Sharpe in 67.6% of market-level cells (median +0.12; 1 significant gain, 0 losses), meta-label + sizing in 67.6% (median +0.12), and position sizing in 70.7% (median +0.05, no significant result either way). They also cut turnover (median Δ -3.46 and -1.90 capital/year). Their gains are small relative to their uncertainty.
- **ML entry is the most harmful component.** It loses in most cells (median -0.15; 8 significant losses) and raises turnover by +10.40 capital/year. The damage is concentrated in the contrarian strategy: 7 of its significant losses are
- **ML exit is strategy-dependent, not uniformly harmful.** For the four trend-following controls its median effect is -0.15, but for RSI(2) it is +0.70, with 6 significant gains — the only component to significantly improve a strategy in several markets (global F2008: +0.78, CI [+0.49, +1.12]). RSI(2)'s fixed ATR profit target caps winners, and a learned continuation exit lets them run; for trend-followers, whose trailing stops already let winners run, the learned exit mostly adds turnover (median +5.36).
- **Independent gates do nothing reliable.** The ML regime filter (median -0.01, 0 significant results) and the rule-based ADX gate (median -0.01, 1 significant loss) are indistinguishable from zero.

Component attribution across the universe. Market-level tests: every strategy $\times$ market $\times$ freeze. Sig. +/- = BH-FDR $q < 0.05$ within the family. Freeze agreement = share of strategy $\times$ market cells whose Δ SR has the same sign in both independent freezes.

Component	Type	Tests	Δ SR > 0	Median Δ SR	Sig. +	Sig. -	Global median	Global > 0	Freeze agree	Δ Turn.
Rule ADX gate	rule	80	46.2%	-0.01	0	1	-0.02	5/10	60.0%	-2.36
ML regime filter	independent	75	45.3%	-0.01	0	0	-0.02	5/10	45.0%	-3.39
ML meta-labeling	narrowing	74	67.6%	+0.12	1	0	+0.08	6/10	66.7%	-4.42
ML position sizing	narrowing	75	70.7%	+0.05	0	0	+0.03	6/10	47.5%	-1.67
ML entry	replacing	75	40.0%	-0.15	0	8	-0.24	2/10	65.0%	+9.54
ML exit	replacing	75	48.0%	-0.03	6	1	-0.17	3/10	65.0%	+4.87
Meta-label + sizing	combination	74	67.6%	+0.12	1	0	+0.08	6/10	66.7%	-2.79

Median market-level ΔSR by strategy and component (significant gains / significant losses at $q < 0.05$ in parentheses).

Strategy	Rule ADX gate	ML regime filter	ML meta-labeling	ML position sizing	ML entry	ML exit	Meta-label + sizing
JMA crossover	-0.42 (0/1)	-0.28 (0/0)	-0.12 (0/0)	+0.05 (0/0)	-0.28 (0/1)	-0.59 (0/1)	-0.11 (0/0)
Dual SMA crossover	+0.01 (0/0)	-0.01 (0/0)	+0.21 (0/0)	+0.04 (0/0)	+0.06 (0/0)	-0.08 (0/0)	+0.21 (0/0)
Donchian breakout	-0.04 (0/0)	+0.00 (0/0)	+0.18 (1/0)	+0.10 (0/0)	-0.06 (0/0)	-0.14 (0/0)	+0.22 (1/0)
Time-series momentum	+0.03 (0/0)	+0.06 (0/0)	+0.09 (0/0)	+0.00 (0/0)	+0.00 (0/0)	-0.11 (0/0)	+0.10 (0/0)
RSI(2) reversion	+0.14 (0/0)	-0.06 (0/0)	+0.20 (0/0)	+0.08 (0/0)	-1.07 (0/7)	+0.70 (6/0)	+0.21 (0/0)

Before vs. after, in full

Table 7 reports every global before/after comparison and Table 8 the full risk profile (CAGR, volatility, Sharpe, Sortino, maximum drawdown, CVaR, turnover, skewness, trades). Two facts frame all ML effects. First, the rule-based controls are weak out of sample: their all-market Sharpe ratios range from -0.39 (RSI(2), F2008) to 0.69 (JMA, F2008), and their walk-forward parameter selection decays sharply out of sample (Table 12; e.g. dual SMA crossover 330 bps per trade in-sample vs. -

14 out-of-sample, time-series momentum 240 vs. -13). The historical edge of these rules is largely absent since 2008 under realistic fills and costs. Second, most drawdown changes are mechanical exposure effects rather than skill: every component that trades less shallows the all-market maximum drawdown in most cases (meta-labeling 10/10, meta-label + sizing 9/10, ML exit 9/10, ML regime filter 9/10; median exposure change -13.1%, -8.1%, -20.2%, -9.8%), whereas ML entry, which raises exposure by +56.7%, deepens it in every case (0/10 shallower). Risk-adjusted return — the Sharpe ratio — is therefore the right yardstick for attribution.

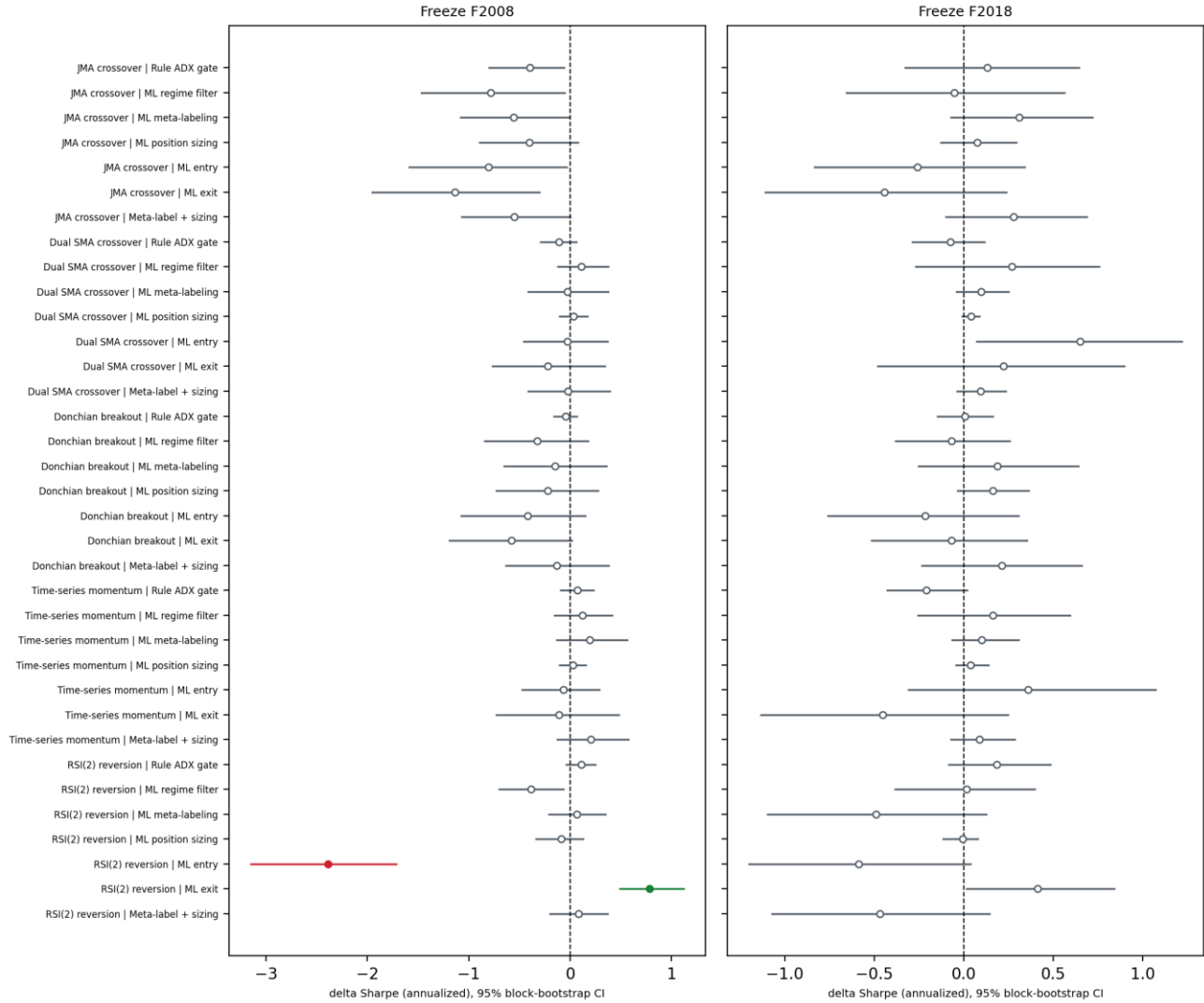
Before vs. after on the all-market portfolio (equal weight across every instrument). ΔSR = annualized Sharpe(variant) – Sharpe(rule-based control); CI = paired circular-block bootstrap (20-day blocks); q = Benjamini–Hochberg across all 70 global tests; $+/–$ = significant gain/loss at $q < 0.05$. $\Delta MaxDD$ in percentage points (positive = shallower); $\Delta Turnover$ in capital/year.

Strategy	Freeze	Component	SR before	SR after	ΔSR	95% CI	q	$\Delta MaxDD$	$\Delta Turn.$
JMA crossover	F2008	Rule ADX gate	0.69	0.29	-0.40	[-0.80, -0.06]	0.257	+0.1%	-2.21
JMA crossover	F2008	ML regime filter	0.69	-0.09	-0.78	[-1.47, -0.06]	0.318	+2.8%	-3.06
JMA crossover	F2008	ML meta-labeling	0.69	0.13	-0.56	[-1.08, -0.00]	0.318	+8.0%	-4.53
JMA crossover	F2008	ML position sizing	0.69	0.28	-0.41	[-0.90, +0.08]	0.432	+1.2%	-1.45
JMA crossover	F2008	ML entry	0.69	-0.12	-0.81	[-1.59, -0.03]	0.318	-9.4%	+5.22
JMA crossover	F2008	ML exit	0.69	-0.44	-1.13	[-1.96, -0.30]	0.117	+4.1%	+21.47
JMA crossover	F2008	Meta-label + sizing	0.69	0.14	-0.55	[-1.07, -0.00]	0.318	+5.7%	-2.24
JMA crossover	F2018	Rule ADX gate	0.51	0.64	+0.13	[-0.33, +0.64]	0.742	+5.2%	-1.54
JMA crossover	F2018	ML regime filter	0.51	0.46	-0.05	[-0.65, +0.56]	0.900	-2.8%	-2.23
JMA crossover	F2018	ML meta-labeling	0.51	0.82	+0.31	[-0.07, +0.72]	0.428	+5.6%	-4.49
JMA crossover	F2018	ML position sizing	0.51	0.59	+0.08	[-0.13, +0.29]	0.661	-12.3%	-1.10
JMA crossover	F2018	ML entry	0.51	0.25	-0.26	[-0.83, +0.34]	0.630	-31.6%	+8.26
JMA crossover	F2018	ML exit	0.51	0.07	-0.44	[-1.11, +0.24]	0.495	-2.6%	+24.84

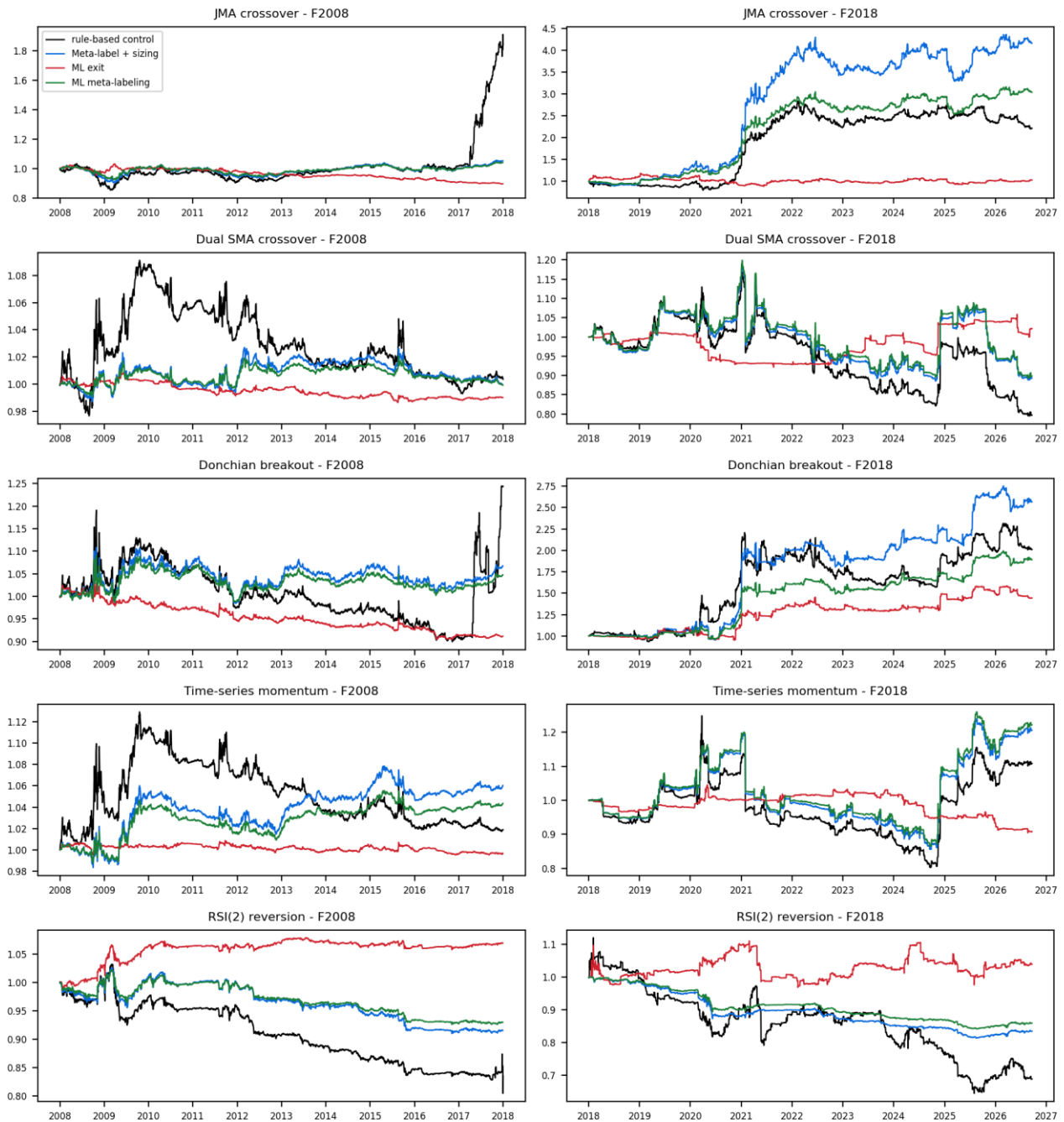
Strategy	Freeze	Component	SR before	SR after	Δ SR	95% CI	q	Δ MaxDD	Δ Turn.
JMA crossover	F2018	Meta-label + sizing	0.51	0.79	+0.28	[-0.10, +0.69]	0.434	-1.0%	-2.23
Dual SMA crossover	F2008	Rule ADX gate	0.03	-0.09	-0.11	[-0.29, +0.06]	0.495	-0.9%	-0.51
Dual SMA crossover	F2008	ML regime filter	0.03	0.14	+0.11	[-0.12, +0.37]	0.616	+5.7%	-1.19
Dual SMA crossover	F2008	ML meta-labeling	0.03	-0.00	-0.03	[-0.42, +0.38]	0.941	+6.9%	-1.96
Dual SMA crossover	F2008	ML position sizing	0.03	0.06	+0.03	[-0.10, +0.17]	0.776	+2.9%	-0.65
Dual SMA crossover	F2008	ML entry	0.03	-0.00	-0.03	[-0.46, +0.37]	0.941	-16.5%	+8.91
Dual SMA crossover	F2008	ML exit	0.03	-0.20	-0.22	[-0.77, +0.34]	0.658	+6.8%	+0.53
Dual SMA crossover	F2008	Meta-label + sizing	0.03	0.00	-0.02	[-0.41, +0.39]	0.949	+6.1%	-1.61
Dual SMA crossover	F2018	Rule ADX gate	-0.15	-0.23	-0.07	[-0.29, +0.12]	0.667	+1.5%	-0.67
Dual SMA crossover	F2018	ML regime filter	-0.15	0.12	+0.27	[-0.27, +0.76]	0.607	+9.0%	-1.66
Dual SMA crossover	F2018	ML meta-labeling	-0.15	-0.05	+0.10	[-0.04, +0.25]	0.434	+6.5%	-1.70
Dual SMA crossover	F2018	ML position sizing	-0.15	-0.11	+0.04	[-0.01, +0.09]	0.428	+2.8%	-0.39
Dual SMA crossover	F2018	ML entry	-0.15	0.50	+0.65	[+0.07, +1.22]	0.257	-5.8%	+9.92
Dual SMA crossover	F2018	ML exit	-0.15	0.07	+0.22	[-0.48, +0.90]	0.739	+22.8%	+0.72
Dual SMA crossover	F2018	Meta-label + sizing	-0.15	-0.06	+0.09	[-0.04, +0.23]	0.434	+6.2%	-1.28
Donchian breakout	F2008	Rule ADX gate	0.27	0.22	-0.04	[-0.16, +0.07]	0.628	-0.7%	-2.47
Donchian breakout	F2008	ML regime filter	0.27	-0.06	-0.32	[-0.85, +0.18]	0.509	+9.7%	-3.52
Donchian breakout	F2008	ML meta-labeling	0.27	0.12	-0.15	[-0.65, +0.36]	0.739	+15.0%	-3.86
Donchian breakout	F2008	ML position sizing	0.27	0.05	-0.22	[-0.73, +0.28]	0.628	+9.5%	-1.35
Donchian breakout	F2008	ML entry	0.27	-0.15	-0.42	[-1.08, +0.15]	0.434	-4.5%	+7.71
Donchian breakout	F2008	ML exit	0.27	-0.31	-0.58	[-1.19, +0.02]	0.366	+11.5%	+14.78
Donchian breakout	F2008	Meta-label + sizing	0.27	0.14	-0.13	[-0.63, +0.38]	0.763	+13.3%	-2.04
Donchian breakout	F2018	Rule ADX gate	0.42	0.43	+0.01	[-0.15, +0.16]	0.952	+0.2%	-2.07
Donchian breakout	F2018	ML regime filter	0.42	0.35	-0.07	[-0.38, +0.26]	0.802	+5.8%	-2.60
Donchian breakout	F2018	ML meta-labeling	0.42	0.61	+0.19	[-0.25, +0.64]	0.628	+14.8%	-3.82
Donchian breakout	F2018	ML position sizing	0.42	0.58	+0.16	[-0.03, +0.36]	0.428	+5.6%	-0.83
Donchian	F2018	ML entry	0.42	0.20	-0.22	[-0.76,	0.628	-30.0%	+10.80

Strategy	Freeze	Component	SR before	SR after	Δ SR	95% CI	q	Δ MaxDD	Δ Turn.
breakout						+0.31]			
Donchian breakout	F2018	ML exit	0.42	0.35	-0.07	[-0.51, +0.36]	0.817	+15.6%	+14.93
Donchian breakout	F2018	Meta-label + sizing	0.42	0.63	+0.21	[-0.23, +0.66]	0.616	+10.1%	-2.07
Time-series momentum	F2008	Rule ADX gate	0.06	0.14	+0.07	[-0.10, +0.23]	0.628	+4.1%	-0.68
Time-series momentum	F2008	ML regime filter	0.06	0.19	+0.12	[-0.16, +0.41]	0.628	+6.8%	-0.80
Time-series momentum	F2008	ML meta- labeling	0.06	0.26	+0.19	[-0.13, +0.56]	0.537	+6.4%	-1.19
Time-series momentum	F2008	ML position sizing	0.06	0.09	+0.03	[-0.11, +0.15]	0.776	+2.9%	-0.46
Time-series momentum	F2008	ML entry	0.06	-0.00	-0.07	[-0.48, +0.29]	0.813	-15.6%	+8.80
Time-series momentum	F2008	ML exit	0.06	-0.05	-0.11	[-0.73, +0.48]	0.810	+8.6%	+3.41
Time-series momentum	F2008	Meta-label + sizing	0.06	0.27	+0.20	[-0.13, +0.57]	0.534	+5.6%	-0.57
Time-series momentum	F2018	Rule ADX gate	0.14	-0.07	-0.21	[-0.43, +0.02]	0.395	+5.3%	-0.76
Time-series momentum	F2018	ML regime filter	0.14	0.30	+0.16	[-0.26, +0.59]	0.682	+15.8%	-0.92
Time-series momentum	F2018	ML meta- labeling	0.14	0.24	+0.10	[-0.06, +0.31]	0.513	+7.8%	-1.14
Time-series momentum	F2018	ML position sizing	0.14	0.18	+0.04	[-0.04, +0.14]	0.601	+6.6%	-0.47
Time-series momentum	F2018	ML entry	0.14	0.50	+0.36	[-0.31, +1.07]	0.601	-2.0%	+10.23
Time-series momentum	F2018	ML exit	0.14	-0.31	-0.45	[-1.13, +0.25]	0.495	+22.2%	+3.87
Time-series momentum	F2018	Meta-label + sizing	0.14	0.22	+0.09	[-0.07, +0.29]	0.566	+7.4%	-0.58
RSI(2) reversion	F2008	Rule ADX gate	-0.39	-0.28	+0.11	[-0.04, +0.25]	0.434	+5.8%	-2.72
RSI(2) reversion	F2008	ML regime filter	-0.39	-0.78	-0.38	[-0.70, - 0.07]	0.192	+8.0%	-2.88
RSI(2) reversion	F2008	ML meta- labeling	-0.39	-0.33	+0.07	[-0.21, +0.35]	0.742	+12.7%	-3.59
RSI(2) reversion	F2008	ML position sizing	-0.39	-0.48	-0.09	[-0.34, +0.13]	0.718	+6.4%	-0.92
RSI(2) reversion	F2008	ML entry	-0.39	-2.78	-2.39 ⁻	[-3.15, - 1.72]	0.000	-40.7%	+22.62
RSI(2) reversion	F2008	ML exit	-0.39	0.39	+0.78 ⁺	[+0.49, +1.12]	0.000	+18.6%	+3.54
RSI(2) reversion	F2008	Meta-label + sizing	-0.39	-0.31	+0.08	[-0.20, +0.37]	0.718	+10.8%	-1.95
RSI(2) reversion	F2018	Rule ADX gate	-0.33	-0.15	+0.18	[-0.08, +0.48]	0.495	+9.6%	-2.79
RSI(2) reversion	F2018	ML regime filter	-0.33	-0.31	+0.02	[-0.38, +0.40]	0.952	+17.7%	-3.14
RSI(2) reversion	F2018	ML meta- labeling	-0.33	-0.82	-0.49	[-1.10, +0.13]	0.434	+26.6%	-5.74

Strategy	Freeze	Component	SR before	SR after	Δ SR	95% CI	q	Δ MaxDD	Δ Turn.
RSI(2) reversion	F2018	ML position sizing	-0.33	-0.34	-0.01	[-0.11, +0.08]	0.996	+3.1%	-0.55
RSI(2) reversion	F2018	ML entry	-0.33	-0.92	-0.59	[-1.20, +0.04]	0.366	-43.5%	+24.88
RSI(2) reversion	F2018	ML exit	-0.33	0.08	+0.41	[+0.02, +0.84]	0.318	+29.6%	+5.24
RSI(2) reversion	F2018	Meta-label + sizing	-0.33	-0.80	-0.47	[-1.07, +0.14]	0.434	+23.8%	-4.70



All-market portfolio: Δ Sharpe with 95% paired block-bootstrap CIs, per strategy and component, in each freeze. Filled = significant at $q < 0.05$.



All-market equity curves in each out-of-sample window: rule-based control vs. meta-labeling, meta-label + sizing, and ML exit.

Full before/after risk profile of the all-market portfolio in each out-of-sample window: CAGR, annualized volatility, Sharpe, Sortino, maximum drawdown, daily CVaR(95%), one-way turnover (capital/year), skewness, trades.

Strategy	Freeze	Variant	CAGR	Vol	SR	Sortino	MaxDD	CVaR	Turn.	Skew	Trades
JMA crossover	F2008	Rule-based control	5.3%	8.0%	0.69	0.87	-17.4%	1.0%	14.01	3.88	11435
		Rule ADX gate	1.6%	6.2%	0.29	0.34	-17.3%	0.7%	11.80	3.16	9677
		ML regime filter	-0.4%	3.5%	-0.09	-0.10	-14.6%	0.5%	10.95	-1.27	9279
		ML meta-labeling	0.3%	2.9%	0.13	0.15	-9.4%	0.5%	9.47	-0.26	8020
		ML position sizing	1.2%	4.4%	0.28	0.33	-16.2%	0.7%	12.56	-0.67	11415

Strategy	Freeze	Variant	CAGR	Vol	SR	Sortino	MaxDD	CVaR	Turn.	Skew	Trades
JMA crossover	F2018	ML entry	-0.9%	5.8%	-0.12	-0.13	-26.8%	0.9%	19.23	-0.60	16333
		ML exit	-0.9%	2.1%	-0.44	-0.49	-13.3%	0.3%	35.48	-1.10	29866
		Meta-label + sizing	0.4%	3.6%	0.14	0.16	-11.7%	0.6%	11.77	-0.28	8020
		<i>Rule-based control</i>	6.5%	14.3%	0.51	0.64	-22.5%	2.0%	14.47	2.74	10189
		Rule ADX gate	7.4%	12.2%	0.64	0.85	-17.3%	1.6%	12.93	3.48	8863
		ML regime filter	4.9%	11.9%	0.46	0.57	-25.3%	1.4%	12.24	2.77	8458
		ML meta-labeling	9.2%	11.5%	0.82	1.06	-16.9%	1.5%	9.98	3.78	7124
		ML position sizing	8.9%	17.0%	0.59	0.72	-34.8%	2.2%	13.37	3.61	10189
		ML entry	3.0%	18.1%	0.25	0.31	-54.1%	2.5%	22.73	1.57	16402
		ML exit	0.2%	8.1%	0.07	0.08	-25.1%	1.2%	39.31	0.54	28299
Dual SMA crossover	F2008	Meta-label + sizing	11.9%	15.9%	0.79	1.01	-23.5%	2.1%	12.24	4.36	7124
		<i>Rule-based control</i>	0.0%	3.0%	0.03	0.03	-9.0%	0.5%	3.09	-0.13	2605
		Rule ADX gate	-0.2%	2.4%	-0.09	-0.10	-9.9%	0.4%	2.58	-0.20	2177
		ML regime filter	0.2%	1.4%	0.14	0.16	-3.3%	0.2%	1.89	-0.16	1573
		ML meta-labeling	-0.0%	1.0%	-0.00	-0.00	-2.1%	0.1%	1.13	0.30	867
		ML position sizing	0.1%	2.2%	0.06	0.06	-6.1%	0.4%	2.44	-0.05	2605
		ML entry	-0.2%	6.3%	-0.00	-0.00	-25.5%	0.9%	12.00	-0.48	10231
		ML exit	-0.1%	0.4%	-0.20	-0.23	-2.2%	0.1%	3.62	0.62	3039
		Meta-label + sizing	-0.0%	1.3%	0.00	0.00	-2.9%	0.2%	1.48	0.23	867
		<i>Rule-based control</i>	-1.8%	9.0%	-0.15	-0.15	-32.0%	1.2%	3.98	-2.27	2873
Dual SMA crossover	F2018	Rule ADX gate	-2.2%	8.4%	-0.23	-0.22	-30.5%	1.1%	3.31	-4.18	2314
		ML regime filter	0.6%	6.8%	0.12	0.14	-23.0%	0.8%	2.31	6.28	1522
		ML meta-labeling	-0.8%	8.6%	-0.05	-0.05	-25.5%	1.1%	2.28	-2.60	1297
		ML position sizing	-1.4%	8.9%	-0.11	-0.11	-29.2%	1.2%	3.59	-2.38	2873
		ML entry	7.5%	17.6%	0.50	0.64	-37.8%	2.4%	13.89	1.92	10170
		ML exit	0.2%	3.3%	0.07	0.07	-9.2%	0.3%	4.69	8.57	3074
		Meta-label + sizing	-0.9%	8.6%	-0.06	-0.06	-25.8%	1.1%	2.69	-2.56	1297

Strategy	Freeze	Variant	CAGR	Vol	SR	Sortino	MaxDD	CVaR	Turn.	Skew	Trades
Donchian breakout	F2008	<i>Rule-based control</i>	1.9%	8.2%	0.27	0.29	-24.5%	1.1%	11.15	-0.57	9241
		Rule ADX gate	1.4%	7.3%	0.22	0.24	-25.1%	0.9%	8.68	-0.44	7151
		ML regime filter	-0.3%	4.1%	-0.06	-0.05	-14.7%	0.6%	7.63	-2.97	6497
		ML meta-labeling	0.4%	4.1%	0.12	0.12	-9.5%	0.6%	7.29	-1.95	6212
		ML position sizing	0.1%	5.1%	0.05	0.05	-15.0%	0.8%	9.80	-1.30	9232
		ML entry	-1.1%	5.8%	-0.15	-0.16	-29.0%	0.9%	18.86	-0.63	16023
		ML exit	-0.8%	2.4%	-0.31	-0.32	-13.0%	0.4%	25.93	-2.65	21946
		Meta-label + sizing	0.6%	4.9%	0.14	0.14	-11.2%	0.7%	9.11	-1.90	6212
Donchian breakout	F2018	<i>Rule-based control</i>	5.7%	16.4%	0.42	0.49	-28.9%	2.1%	11.13	1.23	8295
		Rule ADX gate	5.3%	14.7%	0.43	0.49	-28.7%	1.9%	9.06	1.11	6562
		ML regime filter	3.9%	13.4%	0.35	0.41	-23.1%	1.8%	8.53	1.90	5903
		ML meta-labeling	5.2%	9.0%	0.61	0.66	-14.1%	1.1%	7.31	2.38	5664
		ML position sizing	8.4%	15.9%	0.58	0.71	-23.2%	1.9%	10.30	2.87	8295
		ML entry	2.1%	18.2%	0.20	0.25	-58.9%	2.6%	21.94	1.49	15797
		ML exit	2.9%	9.5%	0.35	0.43	-13.3%	1.1%	26.07	5.09	20220
		Meta-label + sizing	7.7%	13.2%	0.63	0.70	-18.8%	1.5%	9.06	3.74	5664
Time-series momentum	F2008	<i>Rule-based control</i>	0.2%	3.4%	0.06	0.07	-9.9%	0.5%	3.20	-0.69	2714
		Rule ADX gate	0.3%	2.5%	0.14	0.15	-5.9%	0.4%	2.52	-0.41	2139
		ML regime filter	0.3%	1.7%	0.19	0.21	-3.1%	0.3%	2.40	-0.63	2045
		ML meta-labeling	0.4%	1.5%	0.26	0.28	-3.6%	0.2%	2.01	-0.59	1705
		ML position sizing	0.2%	2.7%	0.09	0.10	-7.0%	0.4%	2.74	-0.57	2714
		ML entry	-0.2%	6.3%	-0.00	-0.00	-25.5%	0.9%	12.00	-0.48	10231
		ML exit	-0.0%	0.5%	-0.05	-0.06	-1.3%	0.1%	6.61	1.61	5571
		Meta-label + sizing	0.5%	1.9%	0.27	0.29	-4.3%	0.3%	2.62	-0.65	1705
Time-series momentum	F2018	<i>Rule-based control</i>	0.8%	8.9%	0.14	0.14	-35.8%	1.0%	3.66	-1.29	2886
		Rule ADX gate	-0.8%	7.3%	-0.07	-0.06	-30.5%	0.9%	2.90	-7.74	2314
		ML regime	1.5%	5.3%	0.30	0.33	-19.9%	0.7%	2.74	2.12	2102

Strategy	Freeze	Variant	CAGR	Vol	SR	Sortino	MaxDD	CVaR	Turn.	Skew	Trades
RSI(2) reversion	F2008	filter									
		ML meta-labeling	1.6%	8.1%	0.24	0.23	-27.9%	0.9%	2.52	-2.05	1809
		ML position sizing	1.1%	8.3%	0.18	0.18	-29.2%	1.0%	3.20	-1.83	2886
		ML entry	7.5%	17.6%	0.50	0.64	-37.8%	2.4%	13.89	1.92	10170
		ML exit	-0.8%	2.3%	-0.31	-0.29	-13.5%	0.3%	7.53	-0.31	5941
		Meta-label + sizing	1.5%	8.1%	0.22	0.22	-28.4%	0.9%	3.08	-2.02	1809
		Rule-based control	-1.5%	3.7%	-0.39	-0.40	-22.1%	0.5%	9.51	-2.12	8066
		Rule ADX gate	-1.0%	3.2%	-0.28	-0.28	-16.3%	0.4%	6.80	-3.12	5751
		ML regime filter	-1.3%	1.6%	-0.78	-0.86	-14.1%	0.3%	6.63	-0.86	5587
		ML meta-labeling	-0.6%	1.9%	-0.33	-0.36	-9.4%	0.3%	5.92	-0.71	5022
		ML position sizing	-1.2%	2.5%	-0.48	-0.54	-15.7%	0.4%	8.59	-0.48	8059
		ML entry	-8.0%	3.0%	-2.78	-3.54	-62.8%	0.5%	32.13	2.12	27226
		ML exit	0.6%	1.5%	0.39	0.47	-3.5%	0.2%	13.05	1.35	11026
		Meta-label + sizing	-0.8%	2.3%	-0.31	-0.34	-11.3%	0.4%	7.56	-0.62	5022
RSI(2) reversion	F2018	Rule-based control	-2.9%	8.0%	-0.33	-0.34	-42.4%	1.2%	10.75	-0.48	7666
		Rule ADX gate	-1.3%	7.1%	-0.15	-0.15	-32.8%	1.0%	7.96	-0.25	5564
		ML regime filter	-1.8%	5.4%	-0.31	-0.31	-24.7%	0.8%	7.61	-0.66	5268
		ML meta-labeling	-1.2%	1.5%	-0.82	-0.76	-15.8%	0.2%	5.01	-1.20	4563
		ML position sizing	-2.8%	7.5%	-0.34	-0.33	-39.3%	1.1%	10.20	-0.98	7666
		ML entry	-13.4%	14.6%	-0.92	-1.06	-85.9%	2.4%	35.63	-0.06	26708
		ML exit	0.3%	5.4%	0.08	0.08	-12.8%	0.6%	15.98	1.60	11043
		Meta-label + sizing	-1.4%	1.8%	-0.80	-0.74	-18.6%	0.3%	6.05	-1.23	4563

Replication across freezes

The two freezes are independent replications with different training data and non-overlapping test periods. Agreement is modest (Table 5): the sign of a strategy $\times$ market effect agrees across freezes in 66.7% of cells for meta-labeling, 66.7% for meta-label + sizing, 65.0% for exit and 65.0% for entry, but only 45.0% and 47.5% — no better than chance — for the regime filter and position sizing (cross-freeze correlations of Δ Sharpe: exit 0.70, entry 0.69, meta-labeling 0.48, sizing 0.14). The components with large effects (entry, exit) replicate best;

the small positive effects of the narrowing components replicate only partially.

Generalization across markets

Models fit on every *other* market and applied to a held-out one do at least as well as models fit in-market: transferred meta-labeling has median Δ Sharpe +0.16 (positive in 65.4% of cells, 0 significant losses) and transferred meta-label + sizing +0.15 (65.4%), vs. +0.12 and +0.12 in-market. Whatever the meta-label model learns is not market-specific.

Robustness, stress, and sensitivity

Execution assumptions matter more than any ML component. Filling a breached stop at the stop level instead of at the breaching close — the Study 1 convention — raises the rule-based control’s all-market Sharpe by +0.34 to +0.77 (significant in 10/10 strategy × freeze cases). Under that optimistic fill, ML exit looks far worse (median -0.52, 5/10 significant) than under realistic fills (median -0.17), because the control, not the exit, is flattered; part of Study 1’s “ML exit is harmful” finding is therefore an artifact of optimistic stop fills. A one-day entry lag changes the control by a median -0.04 and leaves the ML comparisons qualitatively unchanged.

Costs. The narrowing components are insensitive to transaction costs (meta-label + sizing: median global Δ Sharpe +0.09, +0.08, +0.08, +0.06 at cost ×0, 1, 2, 4) because they reduce turnover; ML exit’s effect decays from +0.02 to -0.89 (Figure 7).

Budget ablations. Moving the meta-label threshold to 0.45 or 0.55 (median +0.06, +0.08) or shrinking the

Robustness, generalization, and sensitivity. Counts are tests; Sig. columns use BH-FDR within each family.

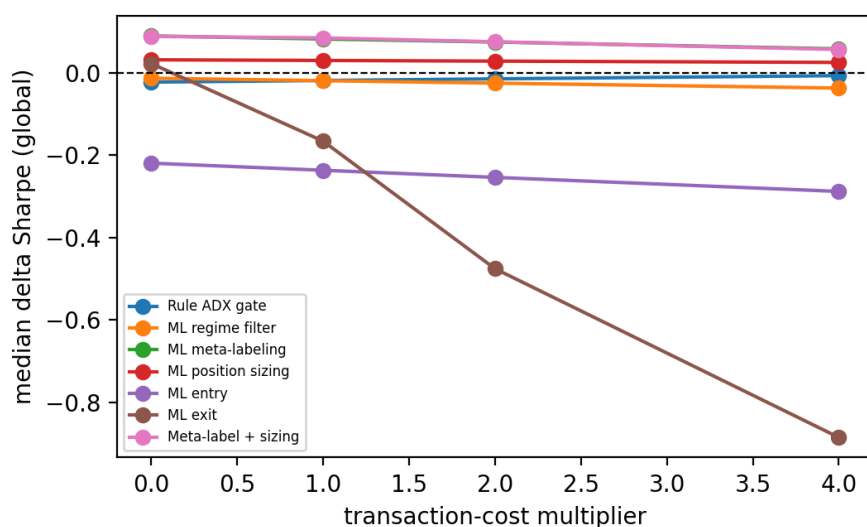
model to 50 trees of depth 3 (meta-labeling +0.19; exit -0.07) leaves the conclusions unchanged: effects come from the insertion point, not from the particular budget.

Regimes and crises. Narrowing components help in low-volatility regimes (meta-labeling +0.38, meta-label + sizing +0.39) and slightly hurt in high-volatility regimes (-0.12, -0.12); ML exit helps in bear markets (+0.38) and hurts in bull markets (-0.42), consistent with cutting trades early (Tables 10, 11).

Survivorship, instruments, markets. Effects in survivorship-free markets and in current-constituent stocks and crypto are similar for the narrowing components (meta-label + sizing median +0.11 vs. +0.14; meta-labeling +0.10 vs. +0.16) and small for exit (+0.03 vs. -0.11), so survivorship does not drive the conclusions. Dropping any single instrument flips the sign of 24.6% of market-level effects — most are fragile — but none of the 8+10 significant ones (0/18 flip); dropping any single market flips 5.7% of global effects.

Test	Variant	Tests	Median Δ SR	Δ SR>			Note
				0	Sig.+	Sig.—	
Cross-market transfer	Meta-label + sizing (fit on other markets)	78	+0.15	65.4%	1	0	in-market +0.12
Cross-market transfer	ML meta-labeling (fit on other markets)	78	+0.16	65.4%	1	0	in-market +0.12
Budget ablation	ML exit, small model	75	-0.07	48.0%	5	3	main -0.03
Budget ablation	Meta-label, small model	74	+0.19	79.7%	3	0	main +0.12
Budget ablation	Meta-label threshold 0.45	75	+0.06	62.7%	1	1	main +0.12
Budget ablation	Meta-label threshold 0.55	72	+0.08	61.1%	6	1	main +0.12
Execution (global)	Rule-based control under one day lag	10	-0.04	30.0%	0	0	vs. realistic control
Execution (global)	Rule-based control under optimistic stop fill	10	+0.41	100.0%	10	0	vs. realistic control
Execution (global)	Meta-label + sizing under one day lag	10	+0.04	60.0%	0	0	vs. control, same execution
Execution (global)	Meta-label + sizing under optimistic stop fill	10	+0.10	80.0%	0	0	vs. control, same execution
Execution (global)	ML exit under one day lag	10	-0.18	40.0%	1	0	vs. control, same execution
Execution (global)	ML exit under optimistic stop fill	10	-0.52	20.0%	1	4	vs. control, same execution
Transaction costs (global)	ML meta-labeling, costs ×0/1/2/4	10	+0.09 / +0.08 / +0.07 / +0.06				median Δ SR per multiplier
Transaction costs (global)	Meta-label + sizing, costs ×0/1/2/4	10	+0.09 / +0.08 / +0.08 / +0.06				median Δ SR per multiplier
Transaction costs (global)	ML exit, costs ×0/1/2/4	10	+0.02 / -0.17 / -0.48 / -0.89				median Δ SR per multiplier
Survivorship	all components, survivorship-free markets	350	+0.03	55.1%	6	4	
Survivorship	all components, survivorship-prone markets	178	+0.05	54.5%	2	6	

Test	Variant	Tests	Median Δ SR	Δ SR>0	Sig.+	Sig.-	Note
Leave-one-out	market tests whose Δ SR sign flips when any one instrument is dropped	528		24.6%			



Median all-market Δ Sharpe by component as transaction costs are scaled.

Cross-regime test: mean Δ SR of the all-market portfolio within regimes defined from information at $t - 1$ (market index vs. its 200-day SMA; 20-day volatility vs. its expanding median). In parentheses: strategy $\times$ freeze cells with Δ SR > 0.

Component	Bull	Bear	Low vol	High vol
Rule ADX gate	-0.05 (4/10)	+0.03 (7/10)	+0.06 (5/10)	-0.07 (4/10)
ML regime filter	-0.06 (5/10)	-0.01 (5/10)	+0.28 (9/10)	-0.19 (3/10)
ML meta-labeling	-0.02 (6/10)	+0.03 (6/10)	+0.38 (8/10)	-0.12 (5/10)
ML position sizing	-0.04 (6/10)	+0.03 (7/10)	+0.20 (9/10)	-0.08 (6/10)
ML entry	-0.60 (2/10)	-0.10 (6/10)	-0.10 (7/10)	-0.45 (2/10)
ML exit	-0.42 (3/10)	+0.38 (6/10)	-0.03 (4/10)	-0.18 (3/10)
Meta-label + sizing	-0.02 (6/10)	+0.03 (7/10)	+0.39 (8/10)	-0.12 (6/10)

Stress test: return and maximum drawdown of the all-market portfolio inside pre-registered crisis windows (mean over strategies).

Crisis	Component	N	Return before	Return after	MaxDD before	MaxDD after
GFC	Rule ADX gate	5	-1.0%	-0.9%	-9.4%	-7.9%
Euro debt crisis	Rule ADX gate	5	-2.2%	-2.1%	-5.4%	-3.9%
2015-16 selloff	Rule ADX gate	5	-1.4%	-1.2%	-4.1%	-3.1%
COVID crash	Rule ADX gate	5	+5.8%	+8.3%	-11.0%	-8.1%
2022 bear market	Rule ADX gate	5	-7.6%	-4.3%	-12.0%	-9.4%
GFC	ML regime filter	5	-1.0%	+0.3%	-9.4%	-5.4%
Euro debt crisis	ML regime filter	5	-2.2%	-1.7%	-5.4%	-3.3%
2015-16 selloff	ML regime filter	5	-1.4%	-1.2%	-4.1%	-2.6%
GFC	ML meta-labeling	5	-1.0%	-0.9%	-9.4%	-4.7%
Euro debt crisis	ML meta-labeling	5	-2.2%	-1.5%	-5.4%	-2.8%
2015-16 selloff	ML meta-labeling	5	-1.4%	-1.0%	-4.1%	-2.4%
GFC	ML position sizing	5	-1.0%	-1.0%	-9.4%	-7.9%
Euro debt crisis	ML position sizing	5	-2.2%	-2.2%	-5.4%	-4.6%
2015-16 selloff	ML position sizing	5	-1.4%	-1.5%	-4.1%	-3.6%
GFC	ML entry	5	-1.0%	+5.9%	-9.4%	-11.9%

Crisis	Component	N	Return before	Return after	MaxDD before	MaxDD after
Euro debt crisis	ML entry	5	-2.2%	-3.4%	-5.4%	-6.8%
2015-16 selloff	ML entry	5	-1.4%	-2.9%	-4.1%	-5.7%
GFC	ML exit	5	-1.0%	+1.1%	-9.4%	-2.8%
Euro debt crisis	ML exit	5	-2.2%	-0.3%	-5.4%	-2.0%
2015-16 selloff	ML exit	5	-1.4%	-0.1%	-4.1%	-1.3%
GFC	Meta-label + sizing	5	-1.0%	-1.1%	-9.4%	-5.7%
Euro debt crisis	Meta-label + sizing	5	-2.2%	-1.8%	-5.4%	-3.5%
2015-16 selloff	Meta-label + sizing	5	-1.4%	-1.3%	-4.1%	-3.0%
COVID crash	ML regime filter	5	+5.8%	+2.7%	-11.0%	-4.1%
2022 bear market	ML regime filter	5	-7.6%	-5.4%	-12.0%	-9.8%
COVID crash	ML meta-labeling	5	+5.8%	+3.1%	-11.0%	-4.1%
2022 bear market	ML meta-labeling	5	-7.6%	-3.9%	-12.0%	-7.4%
COVID crash	ML position sizing	5	+5.8%	+1.9%	-11.0%	-10.4%
2022 bear market	ML position sizing	5	-7.6%	-6.4%	-12.0%	-11.0%
COVID crash	ML entry	5	+5.8%	-1.7%	-11.0%	-16.1%
2022 bear market	ML entry	5	-7.6%	-2.9%	-12.0%	-18.6%
COVID crash	ML exit	5	+5.8%	-0.3%	-11.0%	-4.5%
2022 bear market	ML exit	5	-7.6%	+0.1%	-12.0%	-4.7%
COVID crash	Meta-label + sizing	5	+5.8%	+3.0%	-11.0%	-5.2%
2022 bear market	Meta-label + sizing	5	-7.6%	-4.4%	-12.0%	-8.8%

Overfitting check of the rule-based controls: mean trade return (bps) of the walk-forward-selected parameters in-sample vs. out-of-sample (trade-weighted across markets).

Strategy	In-sample	Out-of-sample	OOS trades
JMA crossover	185.9	45.1	23858
Dual SMA crossover	329.9	-14.4	5934
Donchian breakout	79.7	4.4	19417
Time-series momentum	240.2	-13.5	6020
RSI(2) reversion	14.5	-44.4	17163

Bias audit

Table 13 lists each threat to validity named in the design, the test applied, and its outcome.

Bias audit: each threat to validity, the test applied, and its outcome.

Bias	Test	Outcome
Look-ahead	Truncation invariance of every feature and every strategy signal (unit tests); regimes use $t - 1$ data	pass (tests/test_core.py, tests/test_study2.py)
Data leakage	All models fit once before the freeze; scrambling all post-freeze data leaves every model bit-identical; labels straddling the freeze excluded	pass (tests)
Survivorship	Indices/ETFs/futures/FX analysed separately from current-constituent stocks and crypto	median Δ SR meta-label+sizing: free +0.11 vs. prone +0.14
Selection	Universe, strategies, components, windows, statistics pre-registered in git before data existed; every variant reported	study.toml first committed in 8d9f50f, before all data/results
Overfitting	Walk-forward parameter selection; deflated Sharpe; IS vs. OOS trade-return decay; two independent freezes	see Table 12 and freeze agreement in Table 5
Multiple testing	BH-FDR and Holm within families (528 market-level main tests)	raw $p < 0.05$: 99; BH: 8 gains / 10 losses; Holm: 16
Execution	Close-fill of breached stops (default), 1-day entry lag, cost $\times 0-4$,	Table 9

Bias	Test	Outcome
realism	per-market costs	
Data errors	Two independent sources (FRED) + browser spot-check (MarketWatch)	5-day return corr. 0.82–1.00; spot-check max diff 0.00

Discussion

The taxonomy predicts the variance, not the sign.

Narrowing components — which can only remove or re-weight decisions the rule already makes — produced the smallest and most consistently positive effects in both studies, and in Study 2 not a single significantly negative one. Replacing components produced the largest effects in both directions. What the taxonomy does not predict is *which* direction a replacing component takes: that depends on how well the replaced mechanism fits the strategy. A learned continuation exit is redundant for trend-followers whose trailing stops already let winners run, but it fixes a real weakness of a mean-reversion rule whose fixed profit target truncates them. Symmetrically, a learned entry that finds trends is a poor substitute for a rule designed to fade them. MBCA's practical output is therefore a question to ask of each insertion point: *does the learned model address a known weakness of the rule it replaces?*

What Study 1 got right and wrong. Study 2 confirms Study 1's central conclusion — no component reliably and significantly improves a trend-following control at this budget — and its taxonomy-level reading. It revises three specifics: ML exit is not intrinsically harmful (its Study 1 losses were inflated by optimistic stop fills and are strategy-specific), the Best Combined pairing is positive in most cells but significant in only 1 of 74, and the forex-specific claims do not hold up.

Why the effects are small. The controls themselves have little out-of-sample edge (Table 12), and a matched budget of 8 generic price features gives a learned component little information the rule does not already use. Where the rule has a structural weakness (RSI(2)'s exit), ML finds it; where it does not, the best a narrowing component can do is trade a little less.

Practical implications. For a practitioner with a rule-based system: prefer narrowing components, and expect them to reduce turnover and exposure more than to raise Sharpe; use a replacing component only where the replaced rule has an identifiable weakness, and test it under realistic fills and costs; evaluate with conservative execution before attributing anything to ML; and require replication at a second freeze date before believing a single-window improvement.

Limitations

- **One budget.** Every conclusion is about 8 backward-only price features and one LightGBM configuration. The ablations show the conclusions are stable to the threshold and to a much smaller model, but richer information (fundamentals, order flow, cross-sectional or alternative data) could change them. MBCA is the protocol for testing that fairly, not a claim that the space is exhausted.
- **Daily closes only.** Signals, stops, and fills use daily closes; intraday stop hits are not modelled.

Our close-fill convention is conservative relative to stop-level fills, but overnight gaps and liquidity at the open are not simulated.

- **Data.** Study 2 uses one primary vendor (Yahoo Finance), cross-checked against FRED and MarketWatch. Continuous futures are unadjusted front-month rolls; single stocks are current large caps and crypto the surviving coins (survivorship-prone, analysed separately and found not to drive the conclusions); one planned symbol was unavailable.
- **Portfolio construction.** Equal weighting across instruments with daily mark-to-market is a transparent but simple allocation; volatility targeting or risk parity would change absolute numbers, though not the paired variant-vs-control comparisons.
- **Statistics.** The block bootstrap assumes stationarity within each window; FDR control is applied within pre-defined families, not across the whole study. Most market-level effects are fragile to dropping one instrument, and only the significant ones are shown to be robust to it.
- **Scope.** Five rule families and daily data. MBCA itself is domain-agnostic, but we have not applied it outside trading.

Conclusion

Matched-Budget Component Attribution asks where, specifically, a learned component adds value inside a rule-based system, holding everything but the insertion point fixed. Applied to one strategy in three markets (Study 1) and then, pre-registered, to five historically documented strategies on 153 instruments across 8 markets and two independent freezes (Study 2), it gives a consistent answer: at a matched budget of generic price features, machine learning rarely changes a rule-based strategy significantly (8 significant gains and 10 losses in 528 tests); narrowing components are small, mostly positive, transferable, and safe; replacing components are high-variance and help only when the replaced rule has a structural weakness, as ML exit does for RSI(2) mean reversion; and realistic execution matters more than any component. Because every number in this paper is generated from one pre-registered configuration by the released pipeline, each of these claims can be re-derived — or overturned — by re-running it.

Reproducibility and provenance

Code, data, and this manuscript live in one repository (https://github.com/Cannonbold2412/ML_Component_Attribution). The chain from raw bytes to every number in this paper is:

1. **study.toml** — the pre-registered design (the only configuration the code reads).

2. `python -m mbca study fetch|clean|validate` — raw API responses with URL, UTC timestamp, and SHA-256 (`data/study2/raw/MANIFEST.csv`); cleaned series with per-rule row counts and hashes (`data/study2/clean/MANIFEST.csv`); cross-source reports.
3. `python -m mbca study run` — refuses to start with uncommitted code, configuration, or data; writes every out-of-sample trade and a run manifest (git commit, configuration and data hashes, package versions, thread settings, per-task timings).
4. `python -m mbca study analyze` — all statistics (`results/study2/analysis/*.csv`).
5. `python -m mbca study paper` — every table (`paper/tex/`), figure (`paper/figures/`), and in-text number (`paper/tex/numbers.tex`); the manuscript contains no hand-typed result.

Human decisions and deviations are recorded in `docs/EXPERIMENT_LOG.md`; the git history shows that `study.toml` precedes all Study 2 data and results. Every methodological invariant (no look-ahead in features, strategies, or regime labels; freeze integrity; one insertion point per component; fill realism; MTM accounting; bootstrap size and power) is enforced by the test suite. The Study 1 results in Table 1 are the original study's output files, shipped verbatim.

Errata relative to earlier drafts

No reported Study 1 result changed. (1) The Study 1 significance test was previously described as a paired, instrument-level block bootstrap; the code that produced every Study 1 CI resamples each daily series independently and i.i.d. and compares per-day (non-annualized) Sharpe ratios. It is now described as run, both scales are given, and Study 2 uses a paired block bootstrap instead. (2) The Study 1 forest plots drew the per-day CI around the annualized point estimate; they are regenerated on a common axis. (3) CAGR on a realized-at-exit daily series is not a capital growth rate and is no longer cited for Study 1; Study 2 uses mark-to-market portfolios. (4) An earlier draft stated that ML exit inflates trade count 2.7–4.2×; the source data give 2.5–4.2× (commodities holdout is the minimum). Generating every number from the pipeline removed this class of transcription error.

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